

LECTURE 8
Races and Parallelism

Xuhao Chen
February 17, 2025



Nested Parallelism in Cilk

```
int64_t fib(int64_t n) {  
    if (n < 2)  
        return n;  
    int64_t x, y;  
    cilk_scope {  
        x = cilk_spawn fib(n-1);  
        y = fib(n-2);  
    }  
    return (x + y);  
}
```

The named `child` function may execute in parallel with the `parent` caller

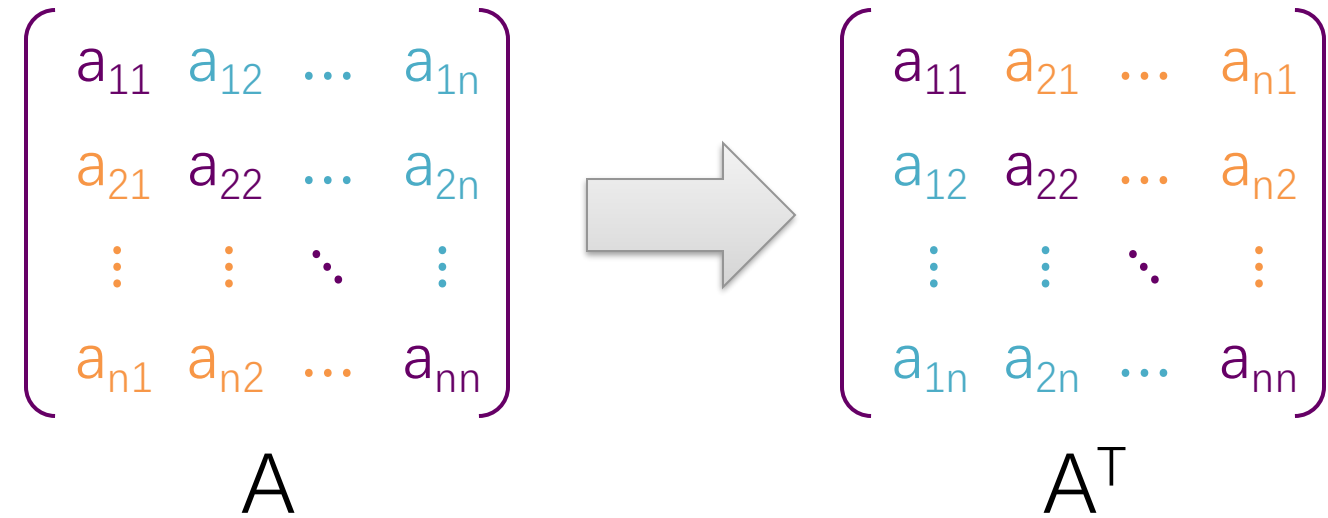
Control cannot exit this context until all spawned children have returned

- Cilk keywords `grant permission` for parallel execution
- They do not `command` parallel execution

Loop Parallelism in Cilk

Example:

In-place **matrix transpose**

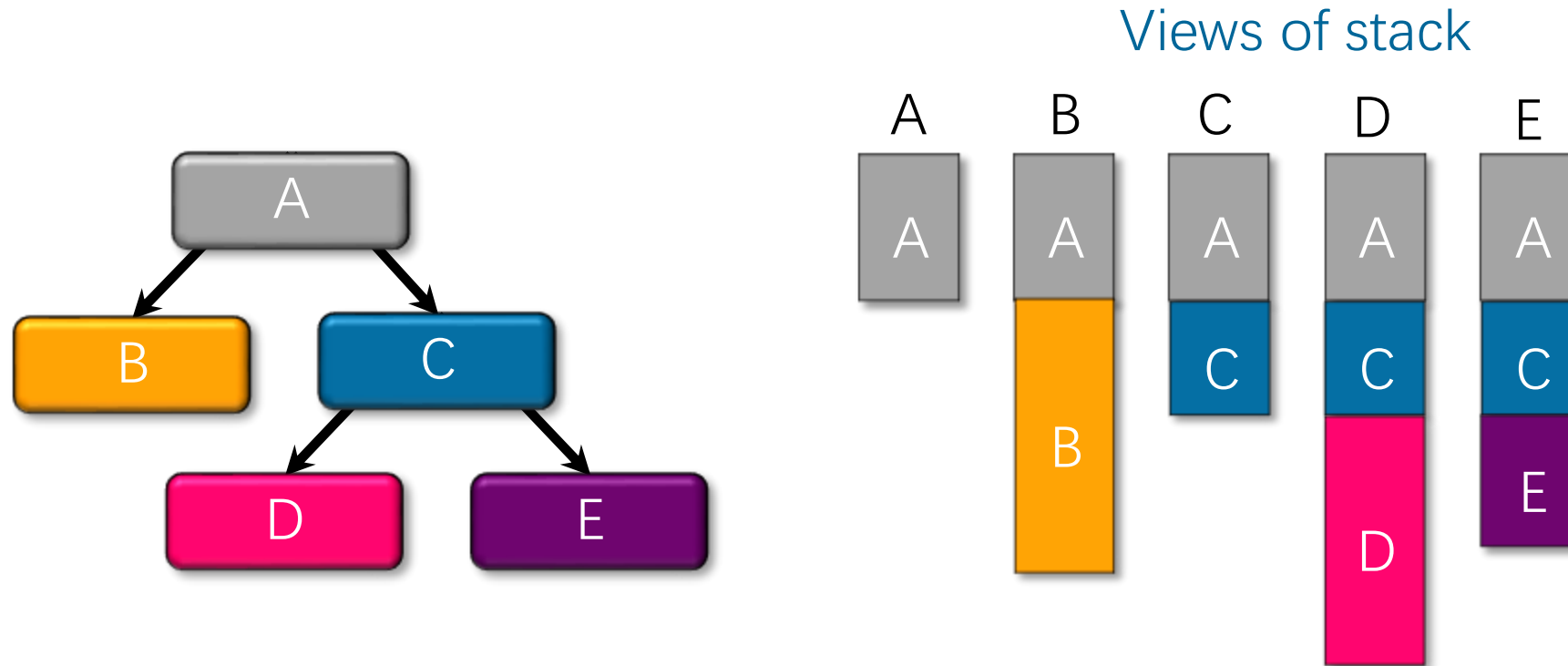


The iterations of a **cilk_for** loop execute in parallel

```
// indices run from 0, not 1
cilk_for (int i=1; i<n; ++i) {
    for (int j=0; j<i; ++j) {
        double temp = A[i][j];
        A[i][j] = A[j][i];
        A[j][i] = temp;
    }
}
```

Cactus Stack

Cilk supports C's [rule for pointers](#): A pointer to stack-allocated memory can be passed from parent to child, but not from child to parent



Cilk's **cactus stack** supports multiple views in parallel

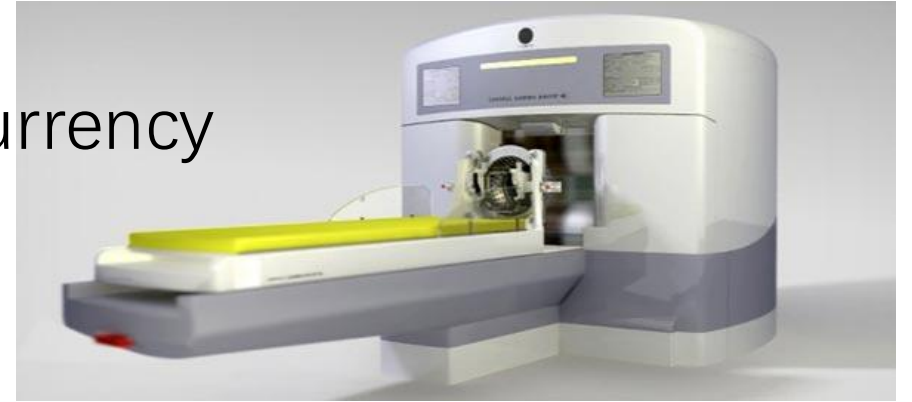
DETERMINACY RACES



Race Conditions

- **Race conditions** are the bane of concurrency
- Famous race bugs include
 - ▣ Therac-25 radiation therapy machine — killed 3 people and seriously injured many
 - ▣ Northeast Blackout of 2003 — left 50 million people without power

Race bugs are notoriously difficult to discover by conventional testing!



Race Condition

- Arise when multiple code paths executing at the same time
- Multiple code paths take a different amount of time than expected
 - ▣ they can finish in a different order than expected



Race Condition

- Arise when multiple code paths executing at the same time
- Multiple code paths take a different amount of time than expected
 - they can finish in a different order than expected
- A **data race** is a type of race condition
 - A C/C++ program containing a data race has **undefined behavior**
- Hard to reproduce and debug as the result is **nondeterministic**

Determinacy Races

Definition. A **determinacy race** occurs when two logically parallel instructions access the same memory location and at least one of the instructions performs a write.

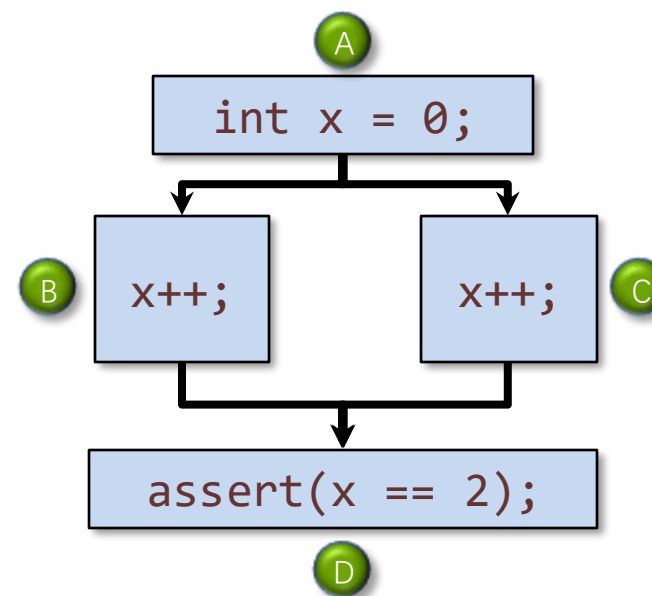
Example

```

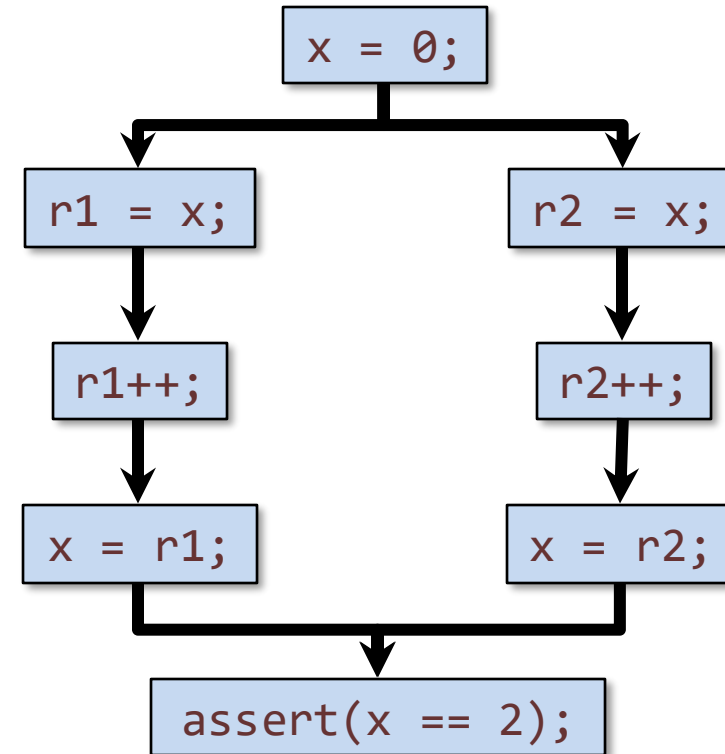
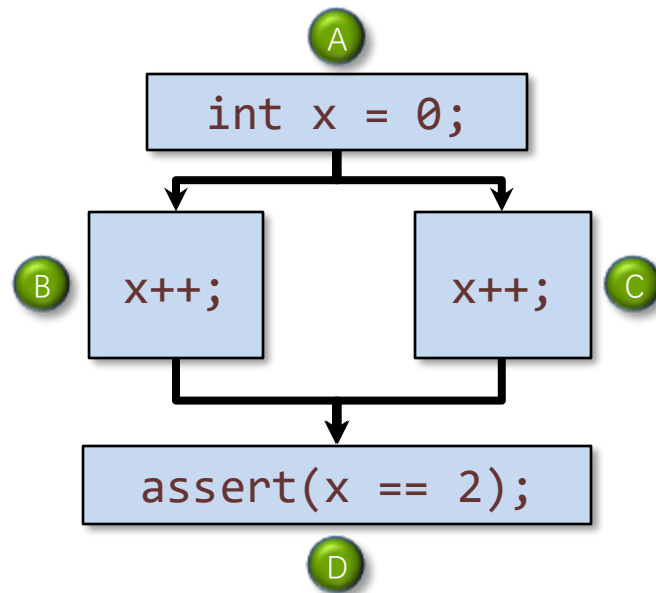
A int x = 0;
  cilk_for (int i=0, i<2, ++i) {
B   x++;
C   }
D assert(x == 2);

```

Trace

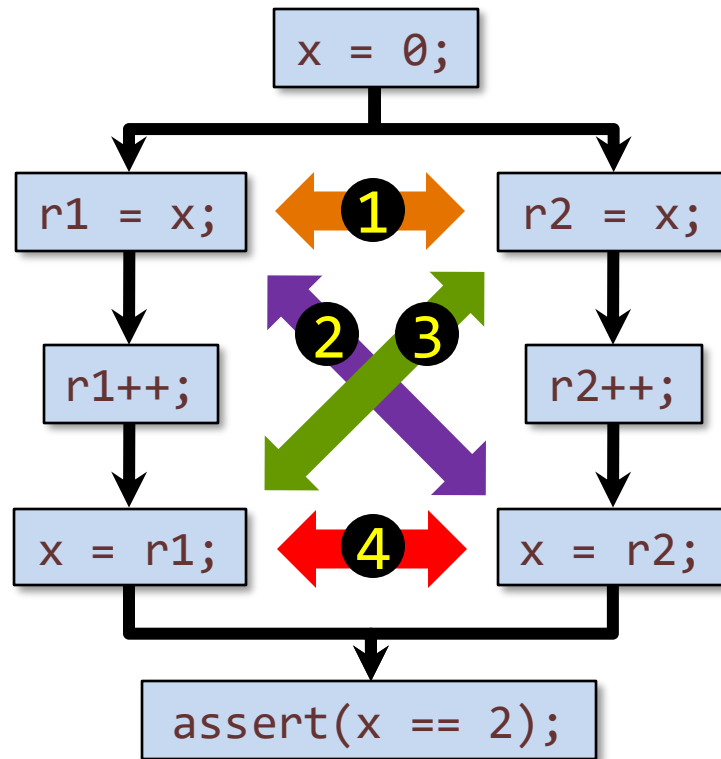


A Closer Look



Race Bugs

Definition. A **determinacy race** occurs when two logically parallel instructions access the same memory location and at least one of the instructions performs a write.

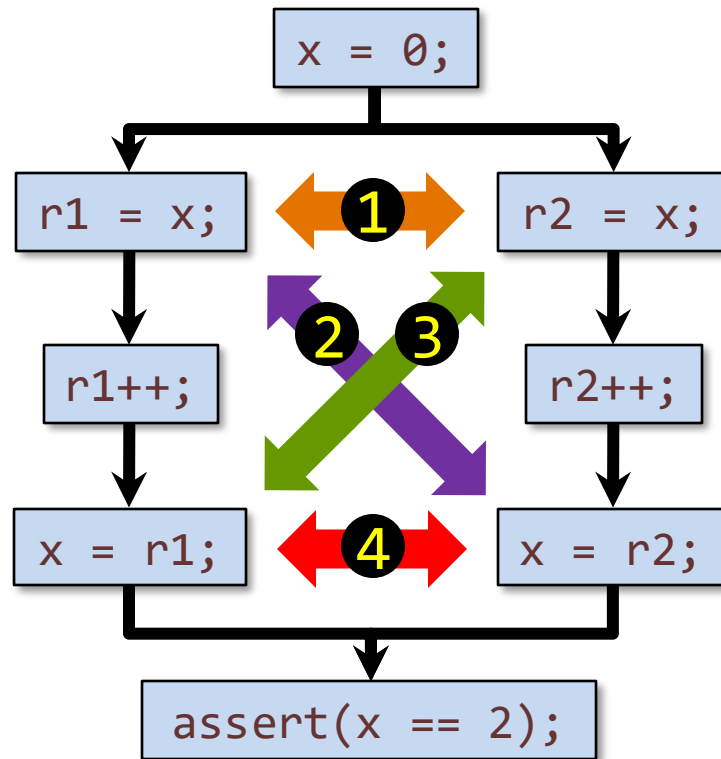


Which of these four pairs of instructions race?

- a. 1
- b. 2
- c. 3
- d. 4
- e. 1 2
- f. 2 3
- g. 3 4
- h. 1 2 3
- i. 2 3 4
- j. None of the above.

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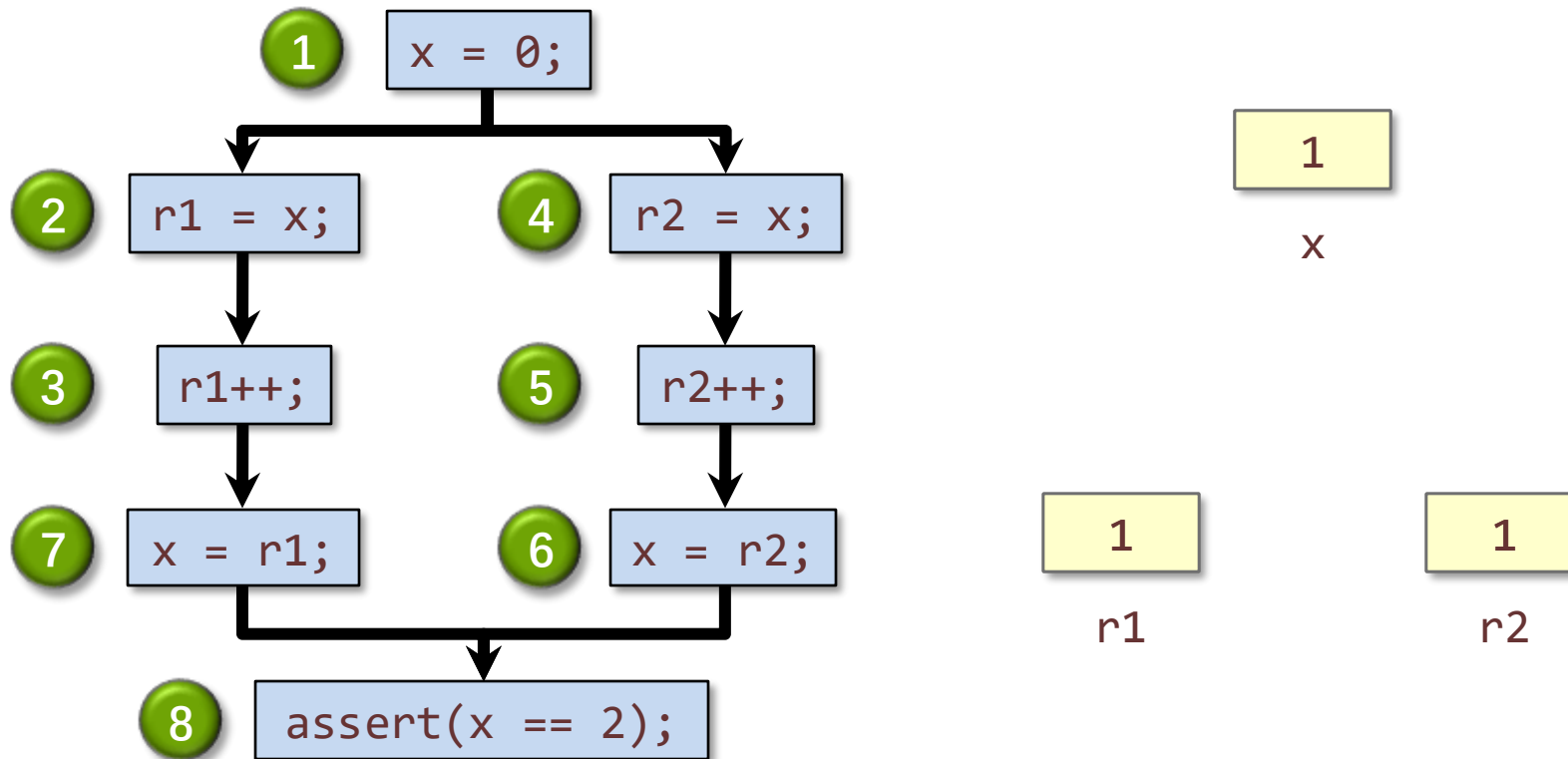


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Race Bugs

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Types of Races

Suppose that instruction **A** and instruction **B** both access a location **x**, and suppose that **A** // **B** (**A** is **parallel** to **B**).

A	B	Race Type
read	read	none
read	write	read race
write	read	read race
write	write	write race

Two sections of code are **independent** if they have no determinacy races between them.

Avoiding Races

- Iterations of a `cilk_for` should be independent.
- After a `cilk_spawn`, the code executed by the spawned task should be independent of the subsequent code executed by the parent and any tasks that the parent spawns or calls, until the `cilk_scope` block is exited
 - arguments to a spawned function are evaluated in the parent before the spawn occurs
- Machine word size matters. Watch out for races in packed data structures:

```
struct {  
    char a;  
    char b;  
} x;
```

Ex. Updating `x.a` and `x.b` in parallel may cause a race! Nasty, because it may depend on the compiler optimization level. (Safe on x86-64)

Cilksan Race Detector

- Compile with `-fsanitize=cilk` to produce a Cilksan-instrumented program
- If there is a determinacy race on a given input, Cilksan **guarantees** to find it
- Cilksan employs a **regression-test** methodology, where the programmer provides test inputs
- Cilksan **identifies** filenames, lines, and variables involved in races, including stack traces
- Ensure that **all** program files are instrumented, or you'll miss some bugs
- Cilksan is your **best friend**.



Race Example: Queens

```
nqueens.c
[...]  
b = (char*) alloca((j+1) * sizeof(char));  
memcpy(b, a, j * sizeof(char));  
for (int i = 0; i < n; i++) {  
    b[j] = i; /* <-- racy write! */  
    if (ok(j+1,b))  
        cnt[i] = cilk_spawn nqueens(n,j+1,b);  
}  
[...]
```

OpenCilk Cilksan Execution

```
...  
b = (char*) alloca((j+1) * sizeof(char));  
memcpy(b, a, j * sizeof(char));  
for (int i = 0; i < n; i++) {  
    b[j] = i; /* <-- racy write! */  
    if (ok(j+1,b))  
        cnt[i] = cilk_spawn nqueens(n,j+1,b);  
}  
...
```

runtime overhead is **nearly constant**
compared with a serial execution

- $\sim 7\times$ slower for this example

```
$ ./nqueens 12  
Running Cilksan race detector  
Running ./nqueens with n = 12.  
Race detected at address 7f7db6c0f2e6  
*   Read 43ef18 nqueens ./nqueens.c:87:3  
|   `--to variable a (declared at nqueens.c:50)  
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|*   Write 43efa9 nqueens ./nqueens.c:89:10  
||   `--to variable b (declared at ./nqueens.c:53)  
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+   Call 43f73b nqueens ./nqueens.c:91:29  
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[...]  
+   Call 43f42b main ./nqueens.c:125:9  
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Stack object b (declared at ./nqueens.c:53)  
Alloc 43eef8 in nqueens ./nqueens.c:86:16  
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2.544000  
Total number of solutions : 14200  
  
Race detector detected total of 1 races.  
Race detector suppressed 3479367 duplicate error  
messages  
$
```

Cilksan Report

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Tips for Effective Performance Engineering

- Maintain the invariant that *your code is correct*.
- *Regression test* heavily and automatically to ensure correctness.
- Don't be a slob: Treat your source code with *respect*.

**Good code hygiene
enables fast code**

WHAT IS PARALLELISM?



Execution Model

```
int64_t fib(int64_t n) {  
    if (n < 2)  
        return n;  
    int64_t x, y;  
    cilk_scope {  
        x = cilk_spawn fib(n-1);  
        y = fib(n-2);  
    }  
    return (x + y);  
}
```

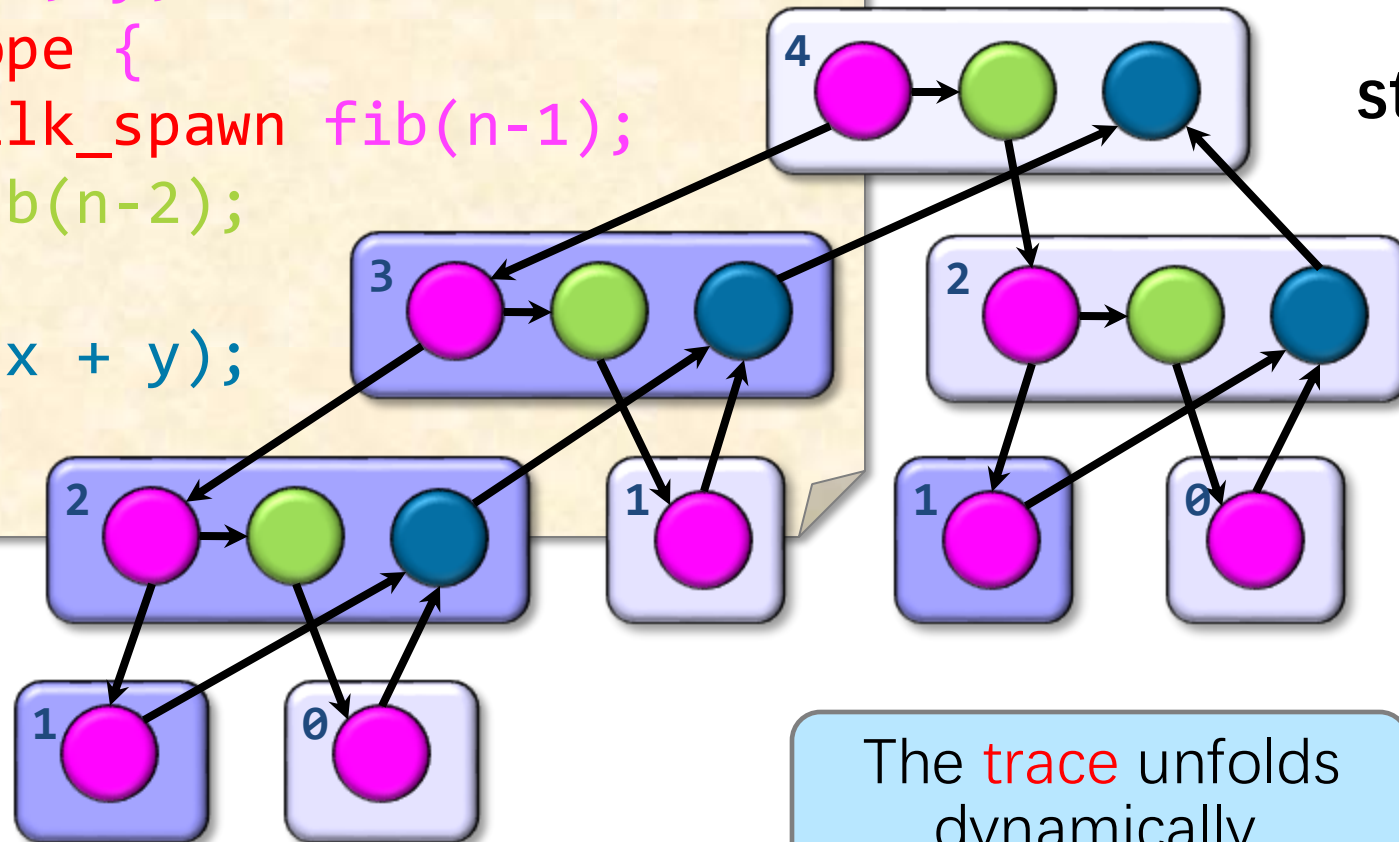
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```

Example:
fib(4)

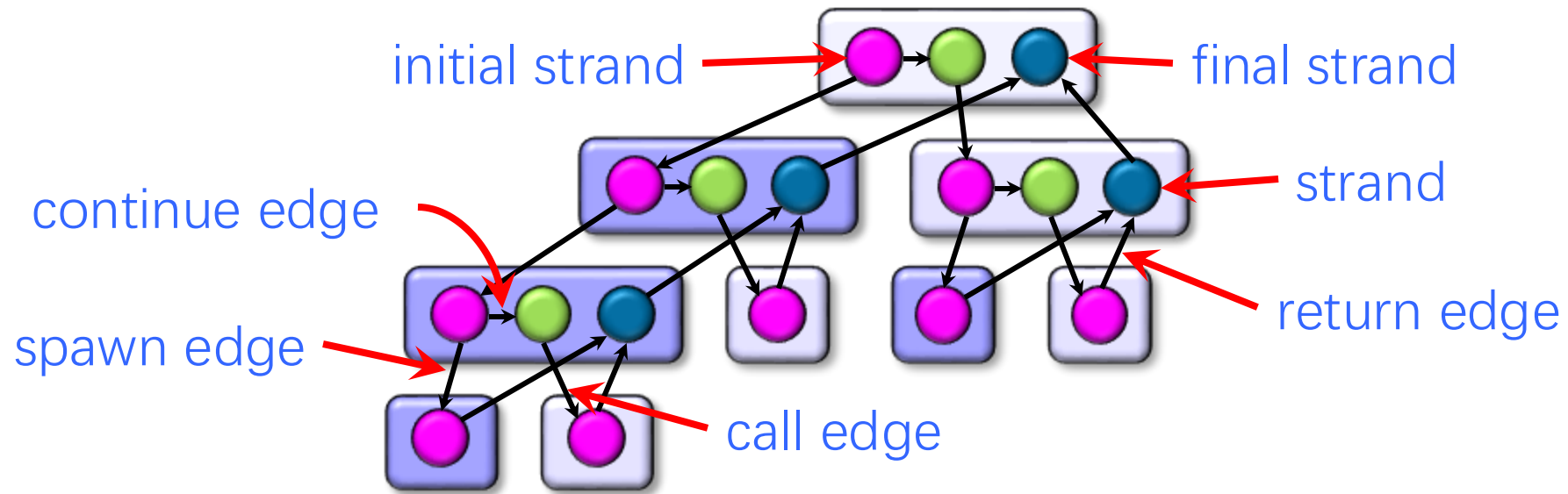
strand

“Processor
oblivious”



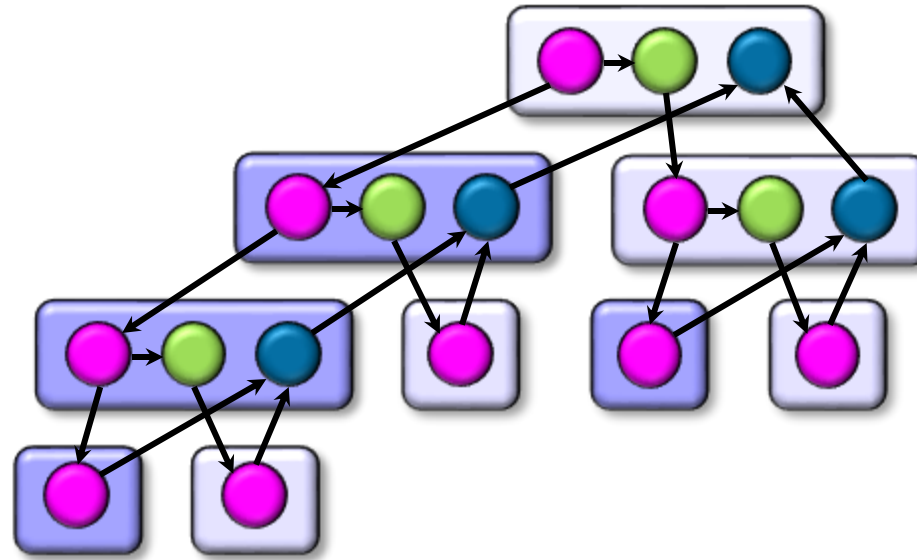
The **trace** unfolds
dynamically.

Trace DAG



- A parallel instruction stream (**trace**) is a dag $G = (V, E)$.
- Each vertex $v \in V$ is a **strand**: a sequence of instructions not containing a spawn, sync, or return from a spawn.
- An edge $e \in E$ is a **spawn**, **call**, **return**, or **continue** edge.
- The compiler converts loop parallelism (**cilk_for**) to spawns and syncs using recursive divide-and-conquer.

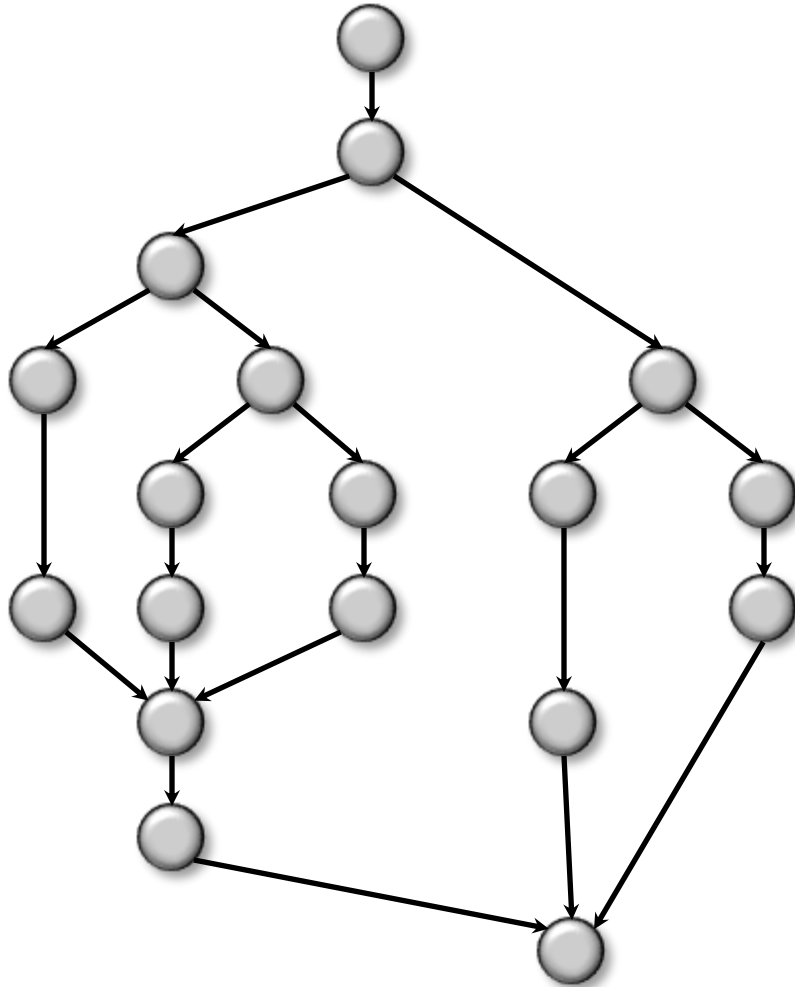
How Much Parallelism?



Assuming that each strand executes in unit time, what is the **parallelism** of this computation?

In other words, what is the **maximum possible speedup** of this computation, where **speedup** is how much faster the parallel code runs compared to the serial code?

Example Trace Dag



Q. What is the **parallelism** (maximum possible speedup) of this computation, assuming that each strand executes in unit time? Pick the closest number.

- a. 1
- b. 2
- c. 3
- d. 4
- e. 5
- f. 6

Amdahl's "Law"



Gene M. Amdahl

If 50% of your application is parallel and 50% is serial, you can't get more than a factor of 2 speedup, no matter how many processors it runs on.*

*Paraphrased.

Amdahl's “Law”

- In general, if a fraction α of an application must be run serially, the speedup can be at most $1/\alpha$.

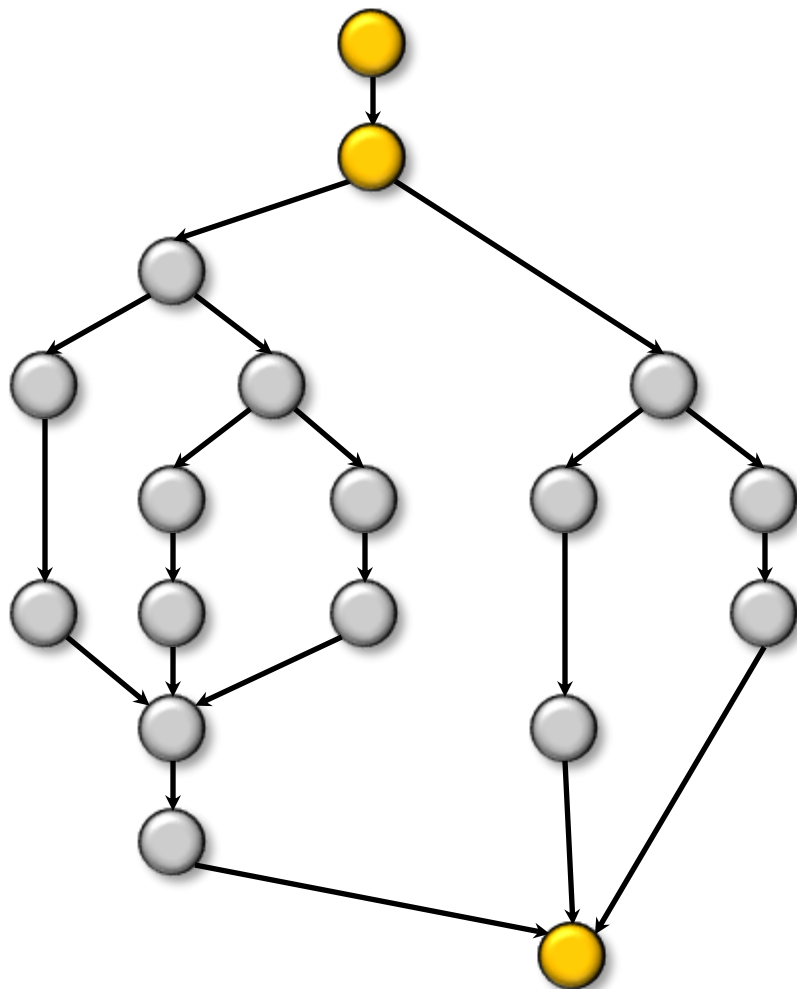
$$\text{Speedup} = \frac{1}{\alpha + \frac{1-\alpha}{P}}$$

α is the a fraction of the application must be run serially

P is the number of processors

Quantifying Parallelism

What is the **parallelism** of this computation?



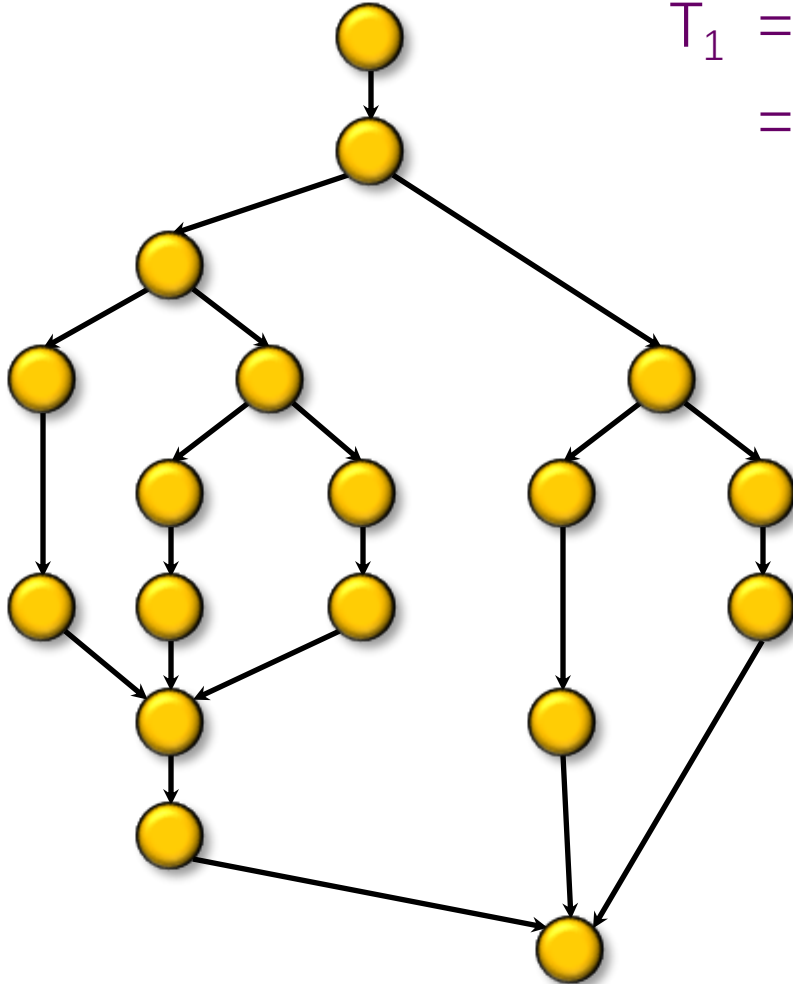
Amdahl's Law says that since the serial fraction is $3/18 = 1/6$, the speedup is upper-bounded by 6.

But this bound is **weak**.

Performance Measures

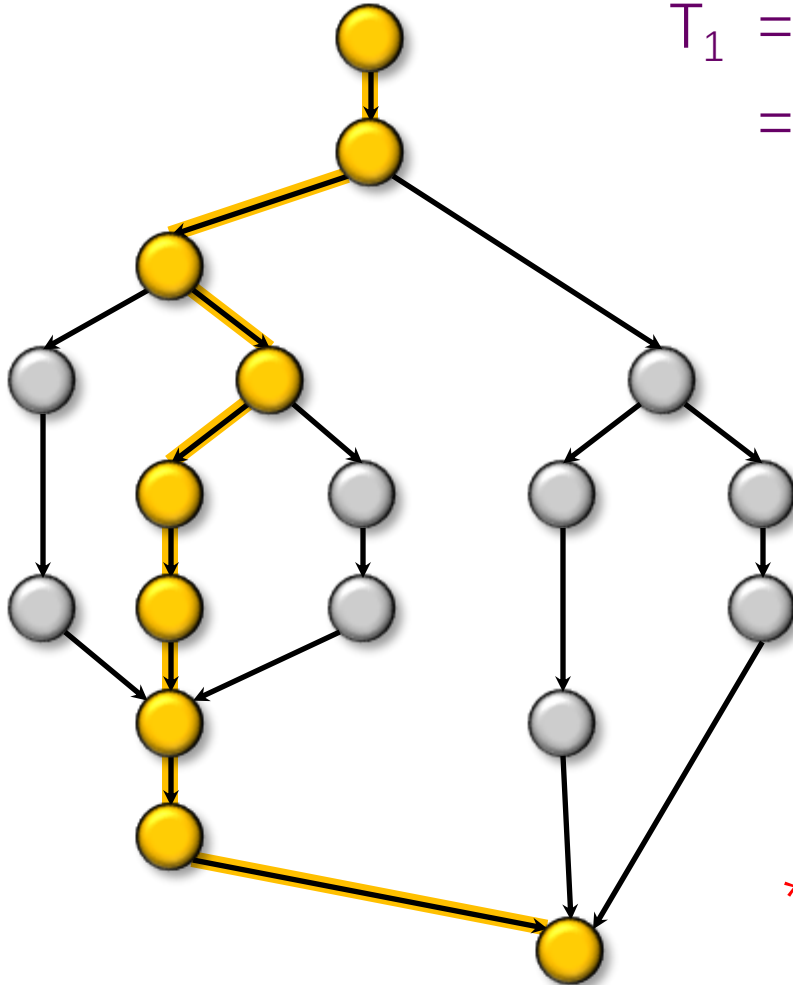
T_p = execution time on P processors

T_1 = work
= 18



Performance Measures

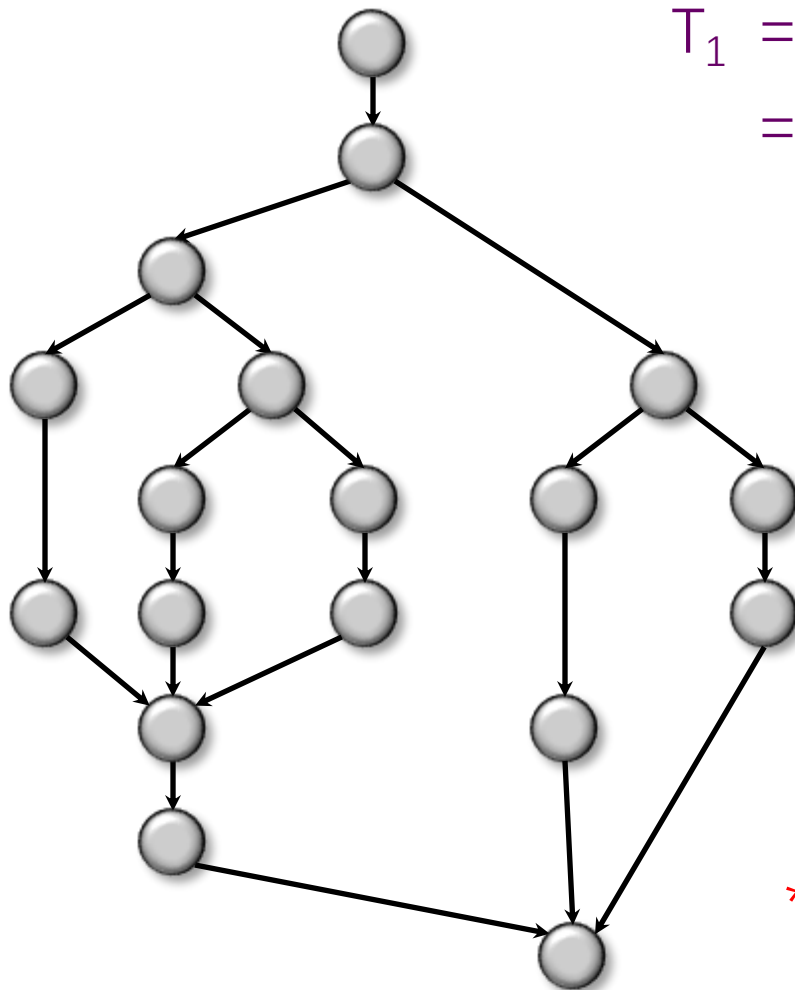
T_p = execution time on P processors

$$T_1 = \text{work} = 18$$
$$T_\infty = \text{span}^* \\ = 9$$


- * Also called critical-path length or computational depth.

Performance Measures

T_p = execution time on P processors



$$T_1 = \text{work} \\ = 18$$

$$T_\infty = \text{span}^* \\ = 9$$

WORK LAW

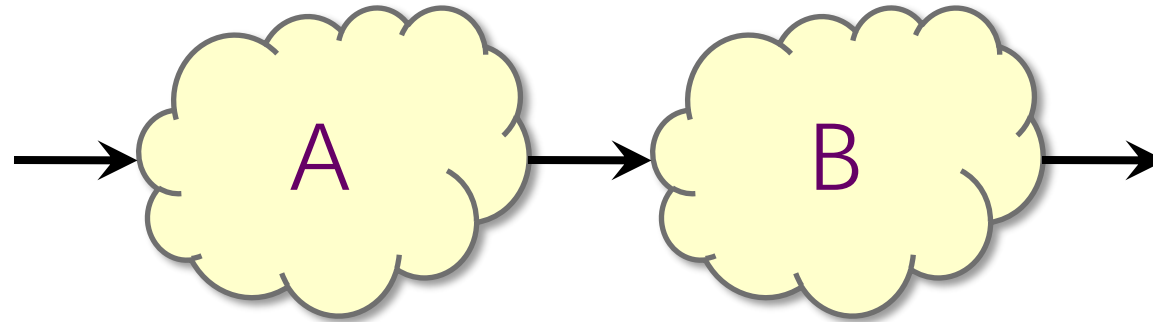
$$\cdot T_p \geq T_1/P$$

SPAN LAW

$$\cdot T_p \geq T_\infty$$

* Also called critical-path length
or computational depth.

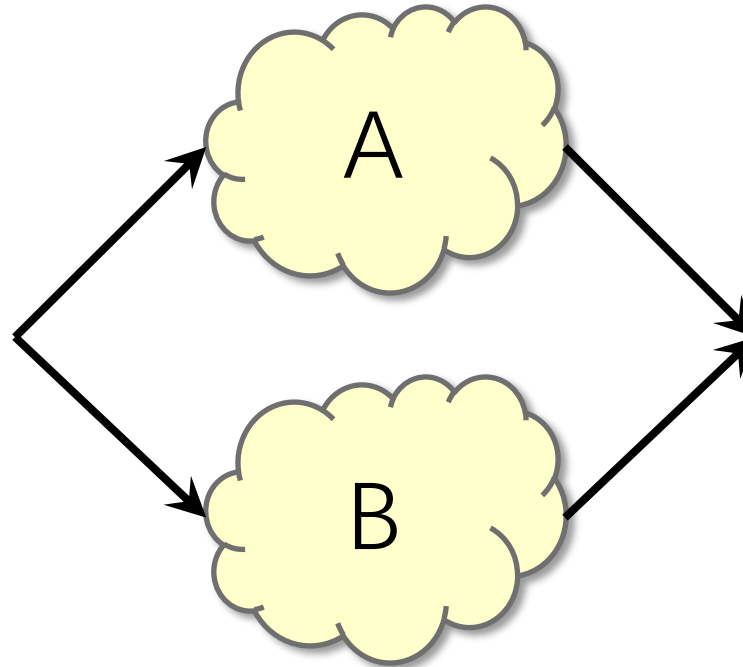
Series Composition



Work: $T_1(A \cup B) = T_1(A) + T_1(B)$

Span: $T_\infty(A \cup B) = T_\infty(A) + T_\infty(B)$

Parallel Composition



Work: $T_1(A \cup B) = T_1(A) + T_1(B)$

Span: $T_\infty(A \cup B) = \max\{T_\infty(A), T_\infty(B)\}$

Speedup

Definition. T_1/T_P = speedup on P processors.

-
- If $T_1/T_P = P$, we have (perfect) linear speedup.
 - If $T_1/T_P < P$, we have sublinear speedup.
 - If $T_1/T_P > P$, we have superlinear speedup, which is not possible in this simple performance model, because of the WORK LAW $T_P \geq T_1/P$.
-

Parallelism

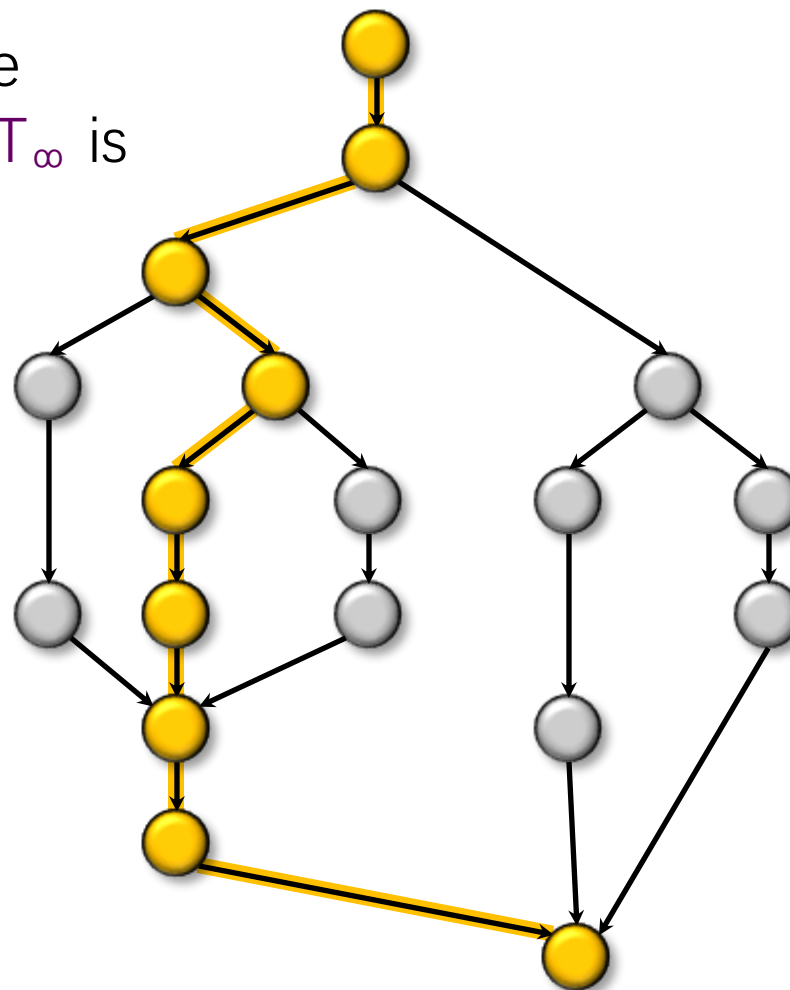
As the **SPAN LAW** dictates that $T_p \geq T_\infty$, the maximum possible speedup given T_1 and T_∞ is

$$T_1/T_\infty = \text{parallelism}$$

= the average amount of work per step along the span

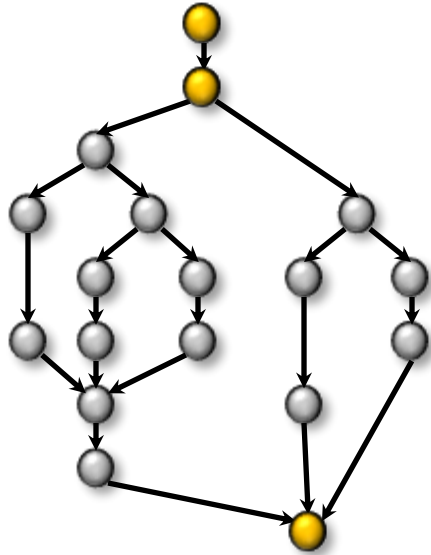
$$= 18/9$$

$$= 2.$$



Amdahl's Law vs. the Span Law

Amdahl's Law for a program where α of T_1 must run serially:



$$T_1/T_\infty = 18/9 = 2$$

$$\text{Speedup} = T_1/T_P = \frac{T_1}{\alpha T_1 + (1-\alpha)T_1/P} = \frac{1}{\alpha + \frac{1-\alpha}{P}}$$

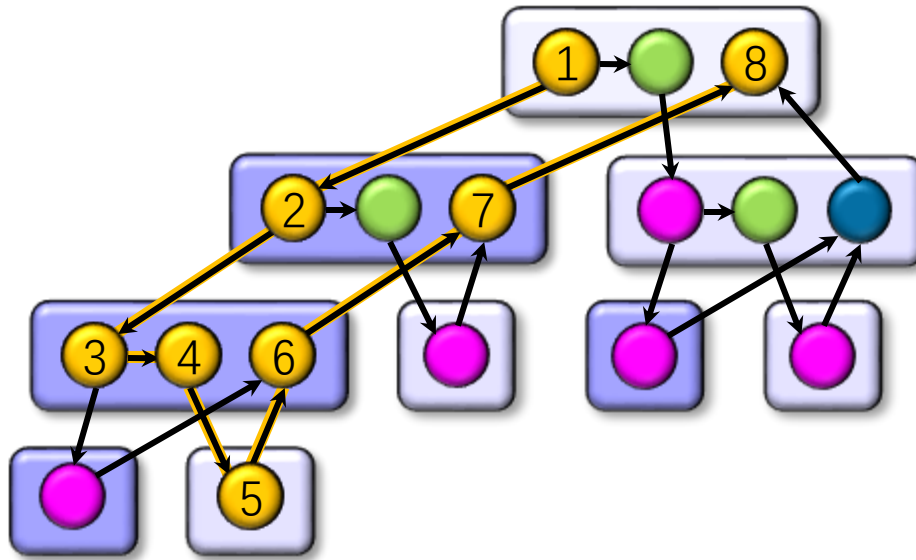
$$P \Rightarrow \infty = \frac{1}{\alpha} = 6$$

$$P = 2 = \frac{18}{3 + 15/2} = 1.7$$

$$P = 10 = \frac{18}{3 + 15/10} = 4.0 \text{ ?}$$

ignores how much of **parallelism** can actually be attained due to **dependencies**

Example: **fib**(4)



Assume for simplicity that each strand in **fib**(4) takes unit time to execute

Work: $T_1 = 17$

Span: $T_\infty = 8$

Parallelism: $T_1/T_\infty = 2.125$

Using many more than **2** processors can yield only marginal performance gains.

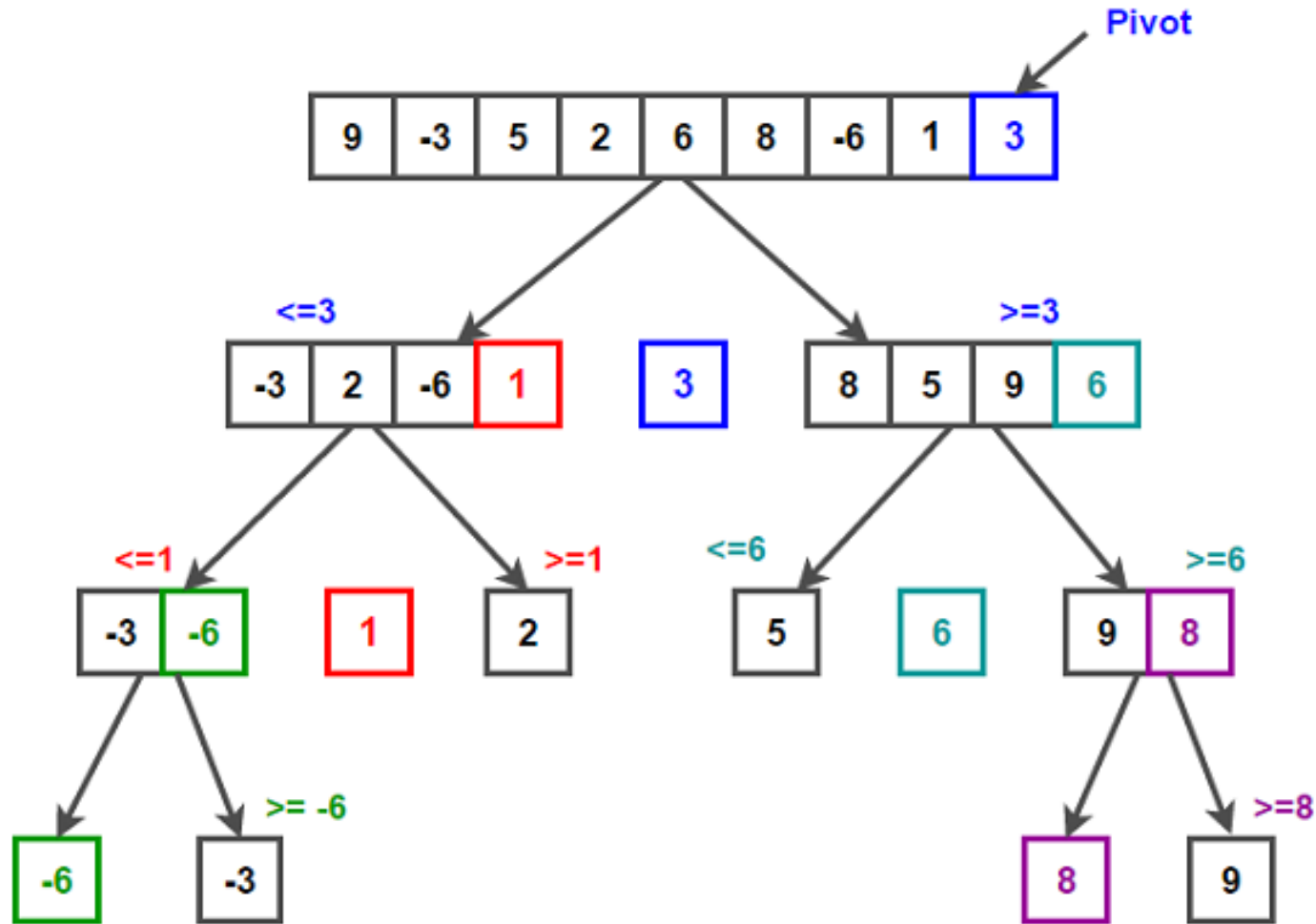
THE CILKSCALE SCALABILITY ANALYZER



Cilkscale Scalability Analyzer

- The OpenCilk compiler provides a scalability analyzer called Cilkscale
- Like the Cilksan race detector, Cilkscale uses compiler instrumentation to analyze a serial execution of a program
- Cilkscale computes work and span to derive upper bounds on parallel performance of all or just part of your program
- Cilkscale is really three tools in one:
 - ▣ an analyzer,
 - ▣ an autobenchmarker,
 - ▣ a visualizer.

Quicksort



Parallelizing Quicksort

Example: quicksort

```
static void qsort(int * begin, int * end)
{
    if (begin < end) {
        int last = *(end - 1);
        // linear-time partition
        int * middle = partition(begin, end - 1, last);
        // move pivot to middle
        swap(end - 1, middle);
        // recurse
        qsort(begin, middle);
        qsort(middle + 1, end);
    }
}
```

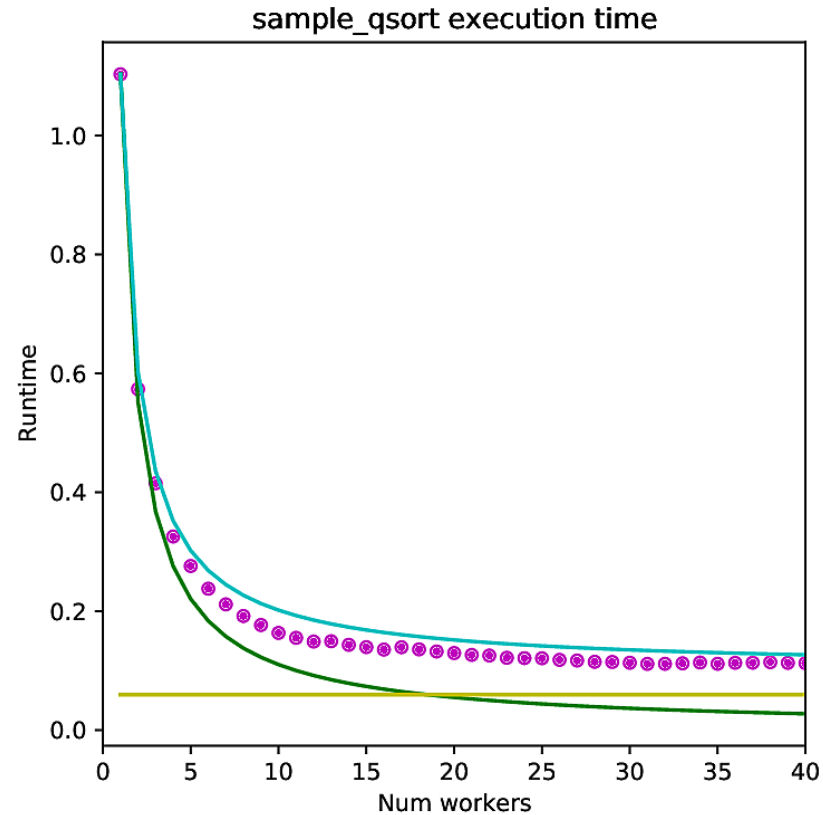
Parallelizing Quicksort

Example: Parallel quicksort

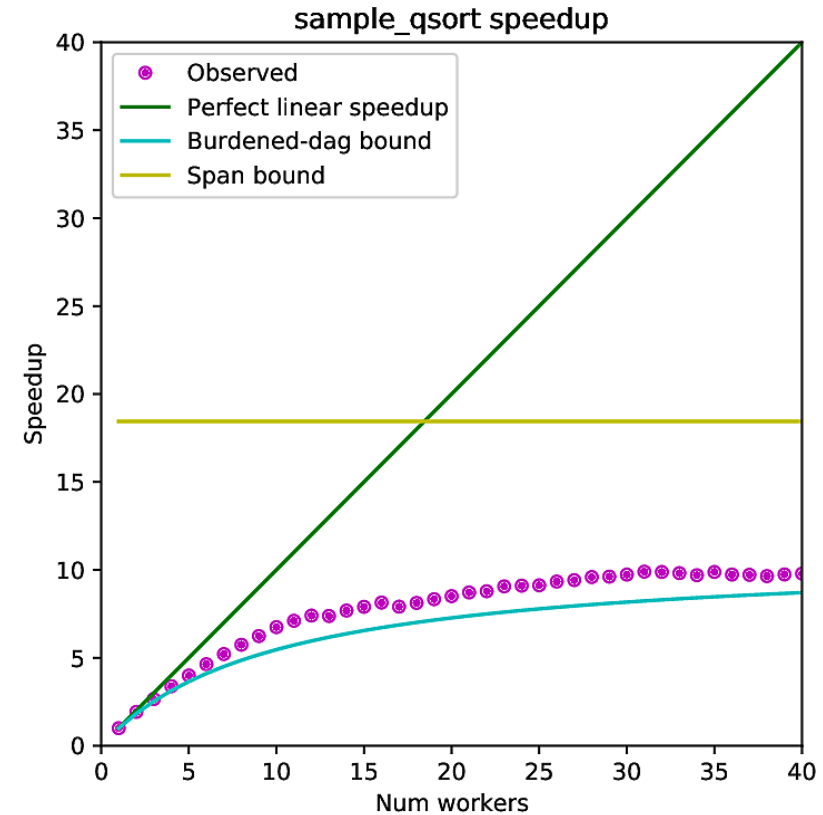
```
static void p_qsort(int* begin, int* end)
{
    if (begin < end) {
        int last = *(end - 1);
        // linear-time partition
        int * middle = partition(begin, end - 1, last);
        // move pivot to middle
        swap(end - 1, middle);
        // recurse
        cilk_scope {
            cilk_spawn p_qsort(begin, middle);
            p_qsort(middle + 1, end);
        }
    }
}
```

Analyze the sorting of 10,000,000 numbers. ★★★ *Guess the parallelism!* ★★★

Cilkscale: Scalability Visualizer

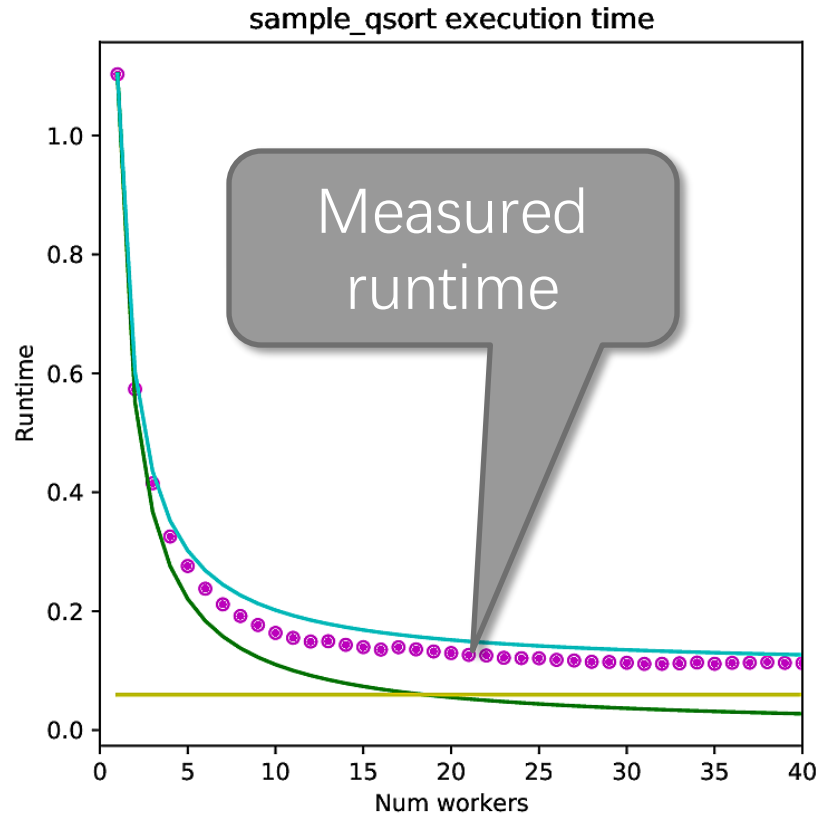


execution time

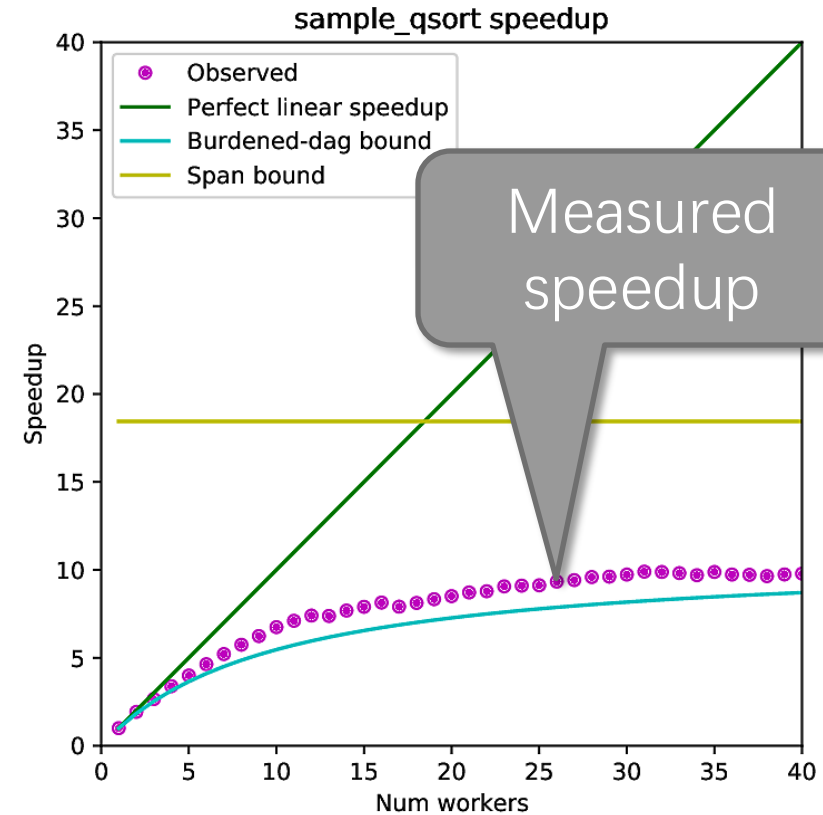


speedup

Cilkscale: Scalability Visualizer



execution time



speedup

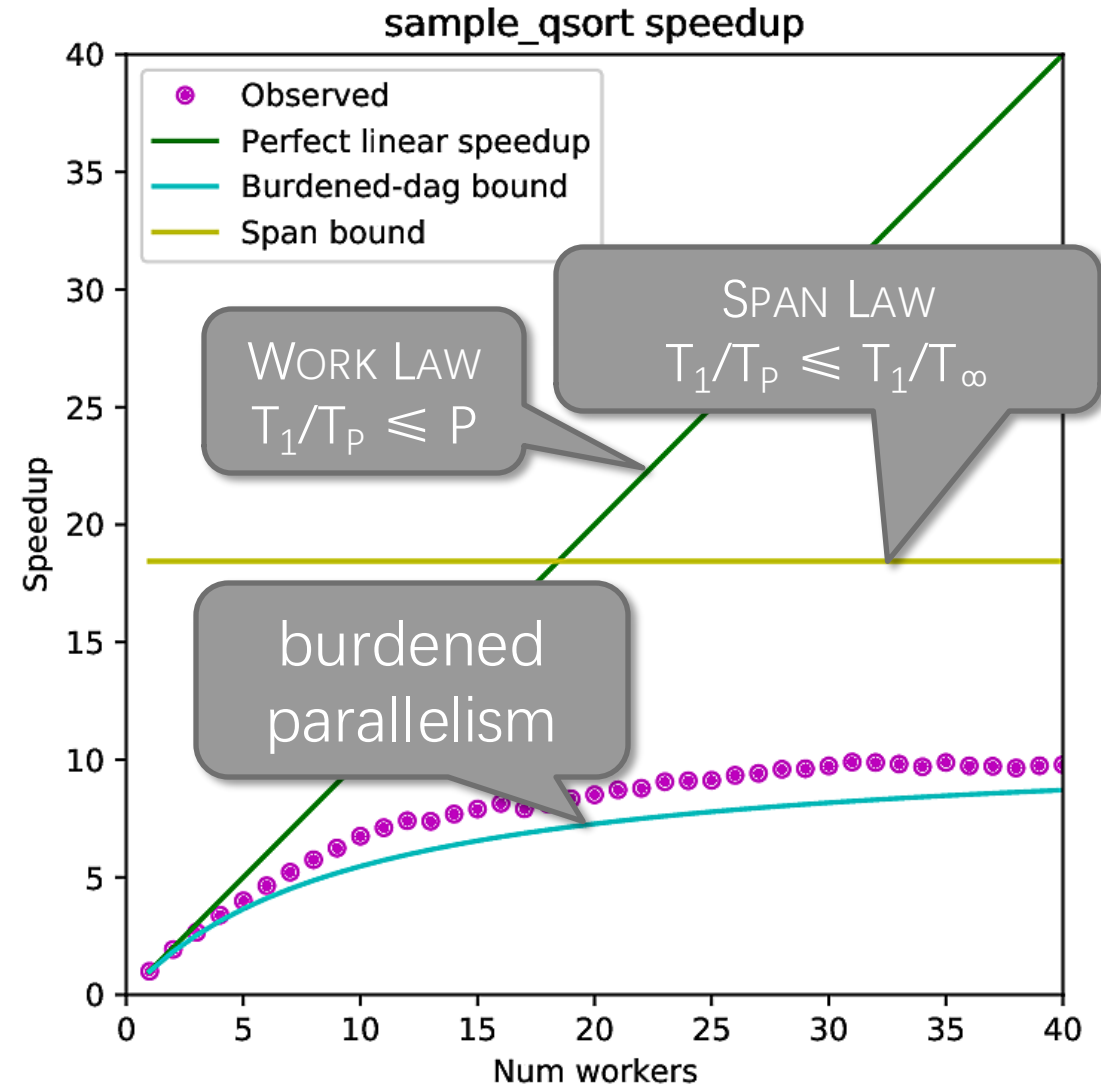
Cilksan **autobenchmarks** the code, running it on 1, 2, 3, ... processors, and the **visualizer** displays the results.

Cilkscale: Speedup Analysis

Cilkscale's **analyzer** determines the **work** and **span**.

The visualizer plots the **WORK** and **SPAN** LAWS.

The visualizer also plots **burdened parallelism**, which indicates whether the program may incur **scheduling overhead**



speedup

Theoretical Analysis

Example: Parallel quicksort

```
static void p_qsort(int* begin, int* end) {  
    if (begin < end) {  
        int last = *(end - 1);  
        // linear-time partition  
        int * middle = partition(begin, end - 1, last);  
        // move pivot to middle  
        swap(end - 1, middle);  
        // recurse  
        cilk_scope {  
            cilk_spawn p_qsort(begin, middle);  
            p_qsort(middle + 1, end);  
        }  
    }  
}
```

Expected work = $\Theta(n \lg n)$

Expected span = $\Theta(n)$

Parallelism = $\Theta(\lg n)$

puny

Interesting Practical* Algorithms

Algorithm	Work	Span	Parallelism
Merge sort	$\Theta(n \lg n)$	$\Theta(\lg^3 n)$	$\Theta(n/\lg^2 n)$
Matrix multiplication	$\Theta(n^3)$	$\Theta(\lg n)$	$\Theta(n^3/\lg n)$
Strassen	$\Theta(n^{\lg 7})$	$\Theta(\lg^2 n)$	$\Theta(n^{\lg 7}/\lg^2 n)$
LU-decomposition	$\Theta(n^3)$	$\Theta(n \lg n)$	$\Theta(n^2/\lg n)$
Tableau construction	$\Theta(n^2)$	$\Theta(n^{\lg 3})$	$\Theta(n^{2-\lg 3})$
FFT	$\Theta(n \lg n)$	$\Theta(\lg^2 n)$	$\Theta(n/\lg n)$
Breadth-first search	$\Theta(E)$	$\Theta(\Delta \lg V)$	$\Theta(E/\Delta \lg V)$

*Cilk on 1 processor competitive with the best C.



Take-Aways



- **Determinacy races** are usually **bugs**.
- Determinacy races can be detected and localized using **Cilksan**
 - ▣ and a good **regression-testing** methodology
- The **WORK & SPAN LAWS** provide lower bounds on the **parallelism**
 - ▣ maximum possible speedup
- **Cilkscale** can analyze the **work**, **span**, and **parallelism** of a computation
- Many **highly parallel** and **work-efficient** algorithms can be programmed in **Cilk**